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FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING (25CS1201E)

2025-26, Term-III

Project Problem Statement Submission Form

Sec No-	**Team No-**	**Project Title-**
Team Lead Roll Number		**2520090018**
Team Lead Name		**Rasagnya M**
Team Member Roll Number		**2520030282**
Team Member Name		**Hasini.P**

1. Abstract

Airlines operate in a highly competitive environment where effective revenue management is essential for profitability. However, analyzing large volumes of operational data—such as ticket sales, passenger load, and route performance—can be complex and time-consuming when not presented in an intuitive format. This creates a need for a system that can simplify data interpretation and support faster, informed decision-making.

The *AeroInsights-Airline Revenue Analytics Dashboard* is a web-based frontend application designed to address this challenge by transforming raw airline data into clear, interactive visual insights. The platform presents key performance indicators (KPIs) such as total revenue, total flights, average ticket price, and load factor, providing a quick overview of operational performance. A dynamic revenue trend chart enables users to monitor changes over time, while load factor visualization highlights seat occupancy efficiency.

In addition, the system includes an insights panel that automatically generates meaningful observations, such as revenue growth trends and potential underperformance. By combining data visualization with user-friendly design, the dashboard allows airline management and analysts to quickly identify patterns, evaluate performance, and make strategic decisions.

Overall, the project demonstrates how modern web technologies can be used to build an efficient analytics tool that enhances business intelligence through clear and interactive data representation.

2. Module Mapping

Module Name	Assigned Roll Number(s)	Signature
Revenue Trend Chart	2520090018	
Load Factor Visualization	2520090018	
Forecast vs Actual Analysis	2520090018	
Insights & Alert Threshold System	2520090018	
Data Integration, State Management & Testing	2520090018	
Authentication & Admin Panel	2520030282	
Dashboard UI & Navigation	2520030282	
Dashboard KPI Cards	2520030282	
Date Range & Filter Controls	2520030282	
Responsive Design & Theme Styling	2520030282	

3. Module Details

— ***Frontend UI & User Interaction***

1. Authentication & Admin Panel

Develops the login/admin interface and manages access to dashboard sections through React Router and LocalStorage.

2. Dashboard UI & Navigation

Designs the overall dashboard structure, including navbar, sidebar, layout organization, and component arrangement.

3. Dashboard KPI Cards

Implements KPI cards displaying total revenue, total flights, average ticket price, and other summary metrics.

4. Date Range & Filter Controls

Develops filtering functionality for selecting date ranges, routes, and comparison periods to dynamically update dashboard data.

5. Responsive Design & Theme Styling

Ensures the application is responsive across devices and applies modern styling using Tailwind CSS / Bootstrap.

— Data Logic, Analytics & Visualization

6. Revenue Trend Chart

Implements interactive revenue trend visualizations using chart libraries such as Recharts or Chart.js.

7. Load Factor Visualization

Develops occupancy and load factor analysis using pie charts, progress indicators, or graphical visualizations.

8. Forecast vs Actual Analysis

Implements comparative analysis between forecasted and actual revenue values through charts and calculated metrics.

9. Insights & Alert Threshold System

Develops logic-based insights and warning notifications for revenue drops, low occupancy, or performance anomalies.

10. Data Integration, State Management & Testing

Handles dataset integration, Axios/API handling, state management using Context API/Redux, and performs application testing and debugging.

4. Frameworks / Platforms / Software Used

Framework / Tool	Category
React.js (v18)	UI Framework
React Router v6	Routing Library
Recharts / Chart.js	Data Visualization
CSS3 / Tailwind CSS	Styling Framework
Context API	State Management
JavaScript ES6+	Core Language
Vite	Build Tool
LocalStorage	Browser API
Mock JSON Data	Data Source
Git & GitHub	Version Control
Netlify / Vercel	Deployment
VS Code	Development Environment

Guide Name and Signature (remarks if any) _____